

Kernel-Agnostic Game-Gradient Position Updates

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1 Allocation identity

Let agent i use a positive differentiable influence kernel $f_\kappa(x_i, b)$ and let

$$G_i(\mathbf{x}, b) = \frac{f_\kappa(x_i, b)}{\sum_j f_\kappa(x_j, b)}, \quad u_i(\mathbf{x}; B) = \mathbb{E}_{b \sim B}[G_i(\mathbf{x}, b)].$$

Writing $s_i(b) = \nabla_{x_i} \log f_\kappa(x_i, b)$ gives the exact identity

$$\nabla_{x_i} u_i = \mathbb{E}_{b \sim B}[G_i(b)(1 - G_i(b))s_i(b)].$$

The score must belong to the configured kernel. Replacing it with $b - x_i$ is valid only for an isotropic Gaussian, up to a constant metric factor.

2 Offline theory versus online observations

The non-naive theory initialization and the σ^* calculation use the offline KDE quadrature

$$B_{\text{KDE}} = \{(b_k, B_k)\}_{k=1}^K, \quad \sum_k B_k = 1,$$

stored as `TraitSpace.grid` and `TraitSpace.weights`. Their gradient is

$$g_i^{\text{off}} = \sum_k B_k G_i(b_k)(1 - G_i(b_k))s_i(b_k).$$

The trained router is online: it does not know the complete resource distribution. In round t it observes only the training batch $\mathcal{B}_t = \{b_{tm}\}_{m=1}^{M_t}$ and uses

$$\hat{g}_i^{(t)} = \frac{1}{M_t} \sum_{m=1}^{M_t} G_i(b_{tm})(1 - G_i(b_{tm}))s_i(b_{tm}).$$

Thus the online rule is the exact gradient of the empirical batch game and a stochastic approximation to the training-resource law. It is intentionally not equal to g_i^{off} on every round. Validation, test, and KDE-grid resources are not used by the normal online update. An offline-KDE update is retained only as an oracle diagnostic.

Soft top- k controls which adapters train on each observed prompt. It does not truncate the sum defining $\hat{g}_i^{(t)}$: all observed prompts and all clone-level allocations enter the position gradient.

3 Constrained simultaneous step

Let P_D be the tangent projector and Π_D the Euclidean projection for the position domain D . The box uses the identity tangent projector. For the probability simplex Δ^{L-1} ,

$$P_\Delta = I - \frac{1}{L} \mathbf{1}\mathbf{1}^\top.$$

One scalar is applied to the complete population gradient:

$$\eta_{\text{eff}} = \min \left\{ \eta, \frac{c}{\max_i \|P_D \hat{g}_i\|_2} \right\}, \quad x_i^+ = \Pi_D(x_i + \eta_{\text{eff}} P_D \hat{g}_i).$$

This cap preserves the joint vector-field direction. There is no per-agent division by $\sum_m G_i(1 - G_i)$ and no inverse-Hessian preconditioner.

4 Implemented kernel scores

Let $d = b - x$, let $C \succ 0$ be the configured shape, and let $r = \sqrt{\delta^2 + d^\top C^{-1} d}$.

Kernel	Position score $s(x, b)$
Gaussian	$C^{-1}(b - x)/\sigma^2$
Hyperbolic	$C^{-1}(b - x)/(\sigma r)$
Dirichlet	$\sigma^{-1}[\log b - \psi(\alpha) + \psi(\alpha_0)\mathbf{1}]$, where $\alpha = \mathbf{1} + x/\sigma$
Product-Beta	$\sigma^{-1}[\log b - \log(1 - b) - \psi(\alpha) + \psi(\beta)]$, where $\alpha = \mathbf{1} + x/\sigma$ and $\beta = \mathbf{1} + (1 - x)/\sigma$

For the Dirichlet kernel $x \in \Delta^{L-1}$ and

$$\alpha_0 - L = \frac{1}{\sigma}, \quad \text{mode Dir}(\alpha) = \frac{\alpha - \mathbf{1}}{\alpha_0 - L} = x.$$

It is therefore mode-parameterized, not mean-parameterized. The common $\psi(\alpha_0)\mathbf{1}$ term is part of the full ambient derivative and vanishes only after projection onto the simplex tangent.

5 Kernel-matched stability

At a symmetric equilibrium x^* found from $P_D \mathbb{E}_B[s(x^*, b)] = 0$, the antisymmetric population mode is

$$A_- = \frac{1}{N^2} \left[(N-1) \mathbb{E}_B[H] + (N-2) \mathbb{E}_B[ss^\top] \right], \quad H = \nabla_x^2 \log f_\kappa.$$

On the simplex the active matrix is $Q^\top A_- Q$, where the columns of Q span $\mathbf{1}^\perp$.

For the hyperbolic family, x^* does not depend on σ . Evaluating at $\sigma = 1$, define the positive log-curvature and score scatter

$$K = -\mathbb{E}_B[H_1(b)], \quad C = \mathbb{E}_B[s_1(b)s_1(b)^\top].$$

After restricting both matrices to the active domain (replace them by $Q^\top K Q$ and $Q^\top C Q$ on the simplex), the supplied closed form is

$$\sigma_{\text{hyp}}^* = \frac{N-2}{N-1} \lambda_{\max} \left(K^{-1/2} C K^{-1/2} \right).$$

The implementation uses a Cholesky reduction of this symmetric generalized eigenproblem. Dirichlet and product–Beta thresholds, whose equilibria move with σ , are instead found by a warm-started logarithmic scan followed by Brent refinement of the first positive-to-negative largest-eigenvalue crossing.

6 Compatibility

An explicit `kernel` block defaults to `position_update: game_gradient`. Configurations without that block continue through the pre-existing Gaussian allocation, centroid update, and analytic threshold implementation unchanged. The historical `theory_matched` centroid is parallel to the isotropic Gaussian gradient on the same resources, but its per-agent normalization gives different simultaneous dynamics and it is not a general-kernel update.